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Procedia Computer Science 00 (2025) 000–000

Procedia Computer Science

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International Conference on Machine Learning and Data Engineering

Dropout-Aware Training for Sensor-Robust Fault Localization with Spatio-Temporal Graph Neural Networks

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Abstract

Spatio-temporal graph neural networks (STGNNs) localize faults accurately in distribution grids instrumented with only a few micro-phasor measurement units (μPMUs), but their evaluations assume every sensor always reports, which deployed systems do not guarantee. Following the STGNN methodology of Karabulut et al. (2026) on the IEEE 123-bus feeder, we train five architectures on our own dataset, the strongest reaching a soft-voting macro-F1 of 0.85, and present what is to our knowledge the first systematic evaluation of STGNN robustness to μPMU loss, dropping $k \in \{0, 1, 2, 5, 10\}$ of 25 sensors at inference. Across the five architectures evaluated here, clean accuracy did not predict robustness: the most accurate one degraded the fastest, falling below 0.10 at 40% loss. The inversion holds in all three seeds but does not reach significance at $n = 3$, so we report it as a consistent observation rather than an established ordering. Dropout-aware training, which masks measurement nodes during training, raises retention at 40% loss from 5.7% to 80.3% on that backbone in simulated RMS phasor data from the IEEE 123-bus feeder ($p_{\text{Holm}} = 0.004$, paired over three training seeds) at measured parity in inference latency and parameter count, and under random loss improves the robustness of all five architectures. Notably, an availability-gated architectural variant only matches it, indicating that sensor-dropout robustness is best addressed during training rather than through the architecture.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

Keywords: Fault localization; Spatio-temporal graph neural networks; Sensor dropout robustness; Missing measurements; Dropout-aware training; Distribution grids

1. Introduction

Rapid and accurate fault localization is essential for automated restoration in modern power distribution grids. As feeders absorb growing shares of distributed energy resources and adopt self-healing schemes, how precisely a

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering.

system identifies where a fault occurred, not merely that one occurred, determines how much of the network can be re-energized before crews are dispatched. Traditional impedance-based and traveling-wave methods assume densely deployed measurement devices, which does not hold when micro-phasor measurement units (μPMUs) are installed at only a few locations because they are expensive and need communication infrastructure, so a fault-localization method must work under partial observability.

Graph neural networks (GNNs) are a natural fit: the feeder's known topology can be encoded so that message passing shares information between connected buses, letting the model infer faults even at buses carrying no sensor. Building on this, [1] introduced a spatio-temporal GNN (STGNN) coupling a GRU temporal encoder with a spatial GNN over a *measured-only* graph in which only instrumented buses are nodes. We adopt their framework and evaluation methodology as the basis for our robustness study.

That work, however, rests on an assumption that does not hold in deployed systems: that every μPMU is always available and functioning. Real sensors drop out through communication loss and hardware failure, and a model trained and evaluated only with a complete sensor set carries no guarantee of degrading gracefully when some inputs go dark. Prior work on incomplete measurement addresses a different regime, assuming a sensor-availability pattern that is fixed and known at training time (Section 2), which an inference-time failure is not. To the best of our knowledge, no previous work has studied how STGNN-based fault localization is affected when sensors fail during inference.

We therefore train the three spatial encoders of [1], with RGSAGE in both mean and max form, which gives four spatial baselines, and their temporal-only GRU reference (Section 3) on our own regenerated dataset under a common protocol (Section 3.3), and make three contributions. First, we quantify how far fault-localization accuracy degrades when $k \in \{0, 1, 2, 5, 10\}$ μPMUs are dropped at inference, across all five architectures. Second, we introduce dropout-aware training, an architecture-agnostic *training method* that randomly masks measurement nodes during training, and show it recovers most of the accuracy lost to sensor dropout. Third, we report a negative result. As an *architectural* counterpart we introduce *SDR-STGNN* (Sensor-Dropout-Robust STGNN), an availability-gated network built on RGSAGE-Max that structurally silences dropped μPMUs; it matches but does not surpass dropout-aware training while costing 18% more inference latency (Section 3.6).

2. Related Work

Early approaches estimate the fault position from circuit physics, either from the one-end impedance [2] or from the arrival time of the fault-induced wavefront [3]; data-driven alternatives, such as the support-vector machines that Livani et al. [4] apply to the IEEE 123-bus feeder, relax the need for an exact circuit model. All three assume measurement coverage dense enough to observe the relevant impedances or wavefronts, and none represents which buses are electrically adjacent. Table 1 positions the closest prior work along the axes that matter here: multi-architecture comparison, sparse observability, evaluation under test-time sensor failure, and mitigation of the resulting loss.

Encoding the feeder as a graph restores the structure that feature-vector models discard. Chen et al. [5] apply a spectral graph convolutional network with Chebyshev filters to the IEEE 123- and 37-bus systems and harden it with a data augmentation against measurement noise and data loss, the closest precedent for the training-time mitigation we study, although it is applied to a static, single-snapshot classifier over the full grid graph. Fan et al. [6] pursue reconstruction instead, pairing a variational graph autoencoder with GraphSAGE to recover latent node information under noise. Both classify a single snapshot, so neither represents how a fault signature evolves over the milliseconds following inception. Spatio-temporal models supply that time axis by placing a temporal encoder ahead of message passing, as in Nguyen et al. [7] and Ngo et al. [8], who combine a one-dimensional CNN with a GCN and with graph attention respectively. This paper builds directly on Karabulut et al. [1], who benchmark RGCN, RGSAGE, and RGATv2 spatial encoders on a shared GRU temporal encoder over the IEEE 123-bus feeder. Their *measured-only* graph promotes only instrumented buses to nodes, so message passing operates over the sensing infrastructure rather than the physical feeder, beating a purely recurrent baseline by up to 11 F1 points at roughly one-sixth of the training time. A companion study [9] instead varies the grid configuration the detector is transferred to.

Zhang et al. [10] confront incomplete measurement directly: a training-free graph Dirichlet imputation reconstructs absent bus measurements by minimizing the graph Dirichlet energy, and a GCN trained with graph contrastive learning then keeps fault signatures separable under the resulting scarcity. This is the most developed treatment of missing data

Table 1. Positioning of the closest prior work (✓ yes, – not studied, ~ partial). "Node loss @ test" denotes whole measurement nodes going dark at inference.

Work	Approach / model	Multi-arch.	Sparse obs.	Node loss @ test	Loss mitigation
Chen et al. (2020) [5]	Spectral GCN (Chebyshev), static	–	~	~	✓
Zhang et al. (2025) [10]	GDIA-GCL: imputation + contrastive	–	✓	~	✓
Karabulut et al. (2026) [1]	STGNN: RGCN, RGSAGE, RGATv2	✓	✓	–	–
This work	**STGNN + node drop-out study**	✓	✓	✓	✓

in this task, and it is instructive for what it assumes: that the set of unmeasured buses is a fixed property of the network, known when both the imputer and the classifier are built.

These lines of work leave the two right-hand columns of Table 1 almost empty: no prior method combines sparse observability with an evaluation of sensors that fail *after* deployment and a mitigation for the accuracy lost when they do; yet on a feeder with 25 μPMUs, which depend on communication links and maintenance, losing a handful is an ordinary operating event.

3. Materials and Methods

3.1. Grid Simulation and Dataset

We generate our own corpus on the IEEE 123-bus distribution feeder, a standard radial benchmark, simulated in OpenDSS through the OpenDSSDirect.py interface. Simulations run in daily (quasi-static time-series) mode at a 1 ms step, so the loads evolve along a predefined profile and each recorded sample is a converged power-flow solution rather than a single operating-point snapshot. This is an RMS, phasor-domain representation: it captures how the voltage envelope evolves across fault inception and over the window that follows, but not the electromagnetic transient itself (traveling waves, DC offset, sub-cycle harmonics), which a quasi-static solver does not model. The fault signature our models learn is therefore the post-inception RMS depression and its spatial pattern across the feeder, and our accuracies should be read as applying to that class of signal. Table 2 gives the full simulation configuration, feature layout, and per-split window counts, every entry computed directly from the generated dataset.

Partial observability is modeled by instrumenting only 25 buses with μPMUs, following the placement of [1] (single-, two-, and three-phase units reporting three-phase RMS voltages): buses 1, 6, 13, 18, 25, 26, 33, 35, 36, 45, 47, 54, 60, 69, 75, 76, 81, 84, 91, 97, 103, 105, 109, 113, and 300, each unit measuring the phases present at its bus in the feeder model. The placement is taken unchanged and is not optimized here; placement optimization is outside the scope of this work. A reduced *measured-only* graph is built from these buses (Fig. 1), with the 25 μPMUs as its nodes. Its edges follow from a breadth-first spanning tree of the feeder rooted at the substation (bus 150): each μPMU is connected to the nearest μPMU upstream of it on that tree, except that a single- or two-phase unit whose nearest upstream unit is also single- or two-phase is not connected there. Two units, at buses 33 and 113, are in that case, and a fallback connects each of them along the shortest feeder path to the μPMU at bus 1, so that no μPMU is left without a neighbor and the graph stays connected; bus 1 is chosen by a fixed implementation rule (the first connected μPMU in bus-number order), not by electrical proximity. The result is a connected tree with 24 edges, so message passing operates on the sensing infrastructure rather than the complete physical feeder. Only 6 of the 25 candidate fault locations coincide with an instrumented bus; the remaining 19 are never directly observed, the central difficulty the spatial model must overcome.

Encoding of missing measurements: Each node carries three voltage channels $[V_1, V_2, V_3]$ and three availability-mask channels $[m_1, m_2, m_3]$ (Table 2). The mask defines what "missing" means throughout this paper: $m_p = 1$ marks an available measurement of phase p, and $m_p = 0$ marks a phase that is not instrumented or a sensor that has dropped out, so a μPMU that fails is encoded exactly like one that was never installed. A dropped sensor has all six channels set to zero. A zero voltage alone is ambiguous, because after z-score normalization it is also a valid reading (the training-mean voltage), so the mask is the only definitive indicator that a node is unobserved. In the clean corpus the mask is static: it encodes the phase instrumentation of each unit, and only 14 of the 25 μPMUs are three-phase, so the

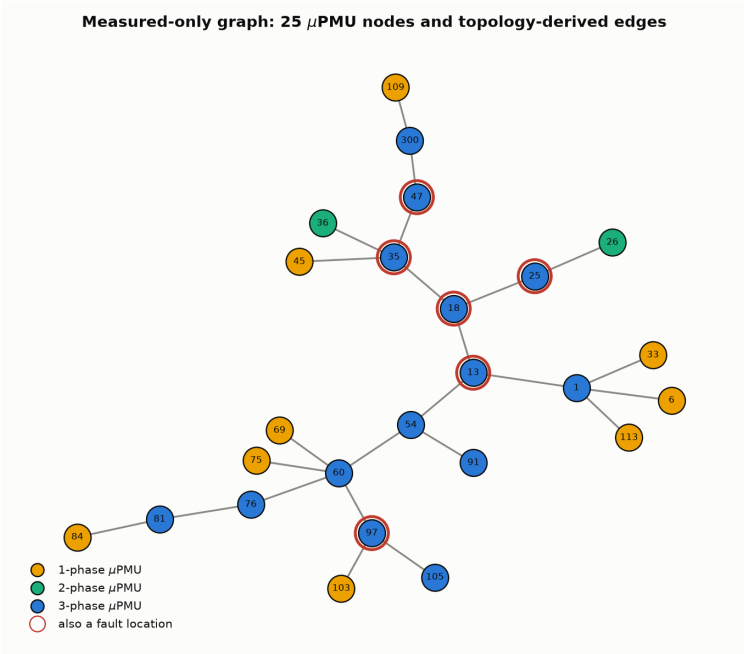


Fig. 1. The measured-only graph of 25 μPMU buses (colored by phase count) and 24 topology-derived edges, with red rings on the 6 buses that are also fault locations. Each edge connects a μPMU to the nearest μPMU upstream of it on the substation-rooted feeder tree, except the two fallback edges 33–1 and 113–1 (Section 3.1).

other units show patterns such as $(1, 0, 0)$ in every window. A standard-trained model therefore never sees the pattern $(0, 0, 0)$, which is the basis of the mechanism in Section 5.2.

The 70/15/15 partition is taken at the *event* level, so the 40 sliding windows of one event never span more than one split and no window of a test event is seen during training. The partition is over events, not over fault scenarios: each of the 825 fault scenarios contributes events to the training and test splits, so the test events are new load realizations of the same 825 fault scenarios. Voltage channels are z-score normalized using training-split statistics only. We adopt the base paper's dataset-generation methodology but build our own corpus (Section 3.3); since every model is trained and evaluated on it with identical splits, the comparisons remain fair. Fault severity varies sharply with fault resistance: for a three-phase-to-ground fault at bus 101, the deepest-sagging μPMU channel drops 0.83 p.u. below its pre-fault level at 0.1 Ω, 0.22 p.u. at 1 Ω, and only 0.01 p.u. at 10 Ω, which is why high-impedance faults are the hardest class to localize throughout this study.

3.2. Baseline Architectures: GRU, RGCN, RGSAGE, RGATv2

We evaluate the three spatial STGNN encoders of [1] (RGCN, RGSAGE, and RGATv2), taking RGSAGE in both its mean- and max-pooling forms so that four spatial baselines result, together with the temporal-only GRU that serves as their non-spatial reference; the GRU has no message-passing stage and is therefore not itself a spatio-temporal GNN. All five follow the designs of [1] unchanged, the one exception being spatial depth, a configuration choice discussed below. They share a common template: a temporal encoder produces one embedding per μPMU node, an optional spatial stage exchanges information over the measured-only graph, and a shared per-node head with soft voting yields the graph-level prediction. Let $x_v \in \mathbb{R}^{S\times F}$ be the input at μPMU node v of the $N = 25$ node measured-only graph, over the $S = 20$ step window with $F = 6$ channels, and let $\mathcal{N}(j)$ denote the neighbors of node j in that graph. A single-layer GRU with 128 hidden units, shared across nodes and applied to each node independently, gives the node embedding of Eq. (1):

Table 2. Dataset configuration and event-level 70/15/15 split of the 50-run corpus.

Simulation configuration	
Test feeder / simulator	IEEE 123-bus, 4.16 kV, 60 Hz; OpenDSS (OpenDSSDirect.py), Daily mode, 1 ms step
Fault buses / μPMUs	25 / 25 (single-, two-, and three-phase units)
Fault types / resistances	11 (3 LG, 3 LL, 3 LLG, 2 three-phase) / 3 (0.1, 1, 10 Ω)
Load multiplier	$\mathcal{U}(0.5,\ 1.3)$ per run
Scenarios × runs	$25 \times 11 \times 3 = 825$ scenarios × 50 runs
Record / windowing	60 ms record (40 ms pre-fault); 20-step (20 ms) windows, 40 per event
Node features F / classes	6 ($[V_1, V_2, V_3]$ + availability mask $[m_1, m_2, m_3]$) / 26 (25 locations + no-fault)
Window-level distribution (event-level split)	
Split	Fault / No-fault / Total windows (events)
Train	577,500 / 577,500 / 1,155,000 (28,875)
Validation	124,000 / 124,000 / 248,000 (6,200)
Test	123,500 / 123,500 / 247,000 (6,175)
Total	825,000 / 825,000 / 1,650,000 (41,250)

$$h_v^{(0)} = \mathrm{GRU}(x_v) \in \mathbb{R}^{128}, \qquad v = 1, \ldots, N. \tag{1}$$

The GRU-only baseline stops at Eq. (1) and cannot exploit spatial relationships. The four GNN variants stack $L = 4$ message-passing layers (hidden dimension 64, ReLU between layers, self-loops added) over that graph and differ only in the aggregation operator: *RGCN* propagates over the symmetrically normalized adjacency (Eq. (2)); *RGSAGE-Mean* and *RGSAGE-Max* concatenate a node's own embedding with a mean- or max-pooled neighborhood before projecting (Eq. (3)) [11]; and *RGATv2* applies $H = 4$ head GATv2 attention over each node's in-neighbors (Eq. (4)) [12], with attention dropout 0.3:

$$h^{(l+1)} = \mathrm{ReLU}\Big(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}\, h^{(l)} W^{(l)}\Big), \tag{2}$$

$$h_j^{(l+1)} = \mathrm{ReLU}\Big(W^{(l)}\Big[h_j^{(l)} \,\Big\|\, \rho(\{\, h_i^{(l)} : i \in \bar{\mathcal{N}}(j)\,\})\Big]\Big), \tag{3}$$

$$h_j^{(l+1)} = \mathrm{ReLU}\left(W_{\mathrm{o}}^{(l)}\Big\|_{c=1}^{H} \sum_{i \in \bar{\mathcal{N}}(j)} \alpha_{ji}^{(l,c)}\, W^{(l,c)} h_i^{(l)}\right), \tag{4}$$

where $\tilde{A} = A + I$ is the adjacency with self-loops and $\tilde{D}$ its degree matrix, $\bar{\mathcal{N}}(j) = \mathcal{N}(j) \cup \{j\}$ is the closed neighborhood (self-loops are added in every variant), $W^{(l)}$ are the layer weights, $W_{\mathrm{o}}^{(l)}$ projects the H concatenated attention heads back to the layer width, $\|$ is concatenation, ρ is mean or max pooling, and $\alpha_{ji}^{(l,c)}$ is the GATv2 attention coefficient of head c. The ReLU separates consecutive hops and is omitted after the last one, whose output passes directly to the classifier head. These operators are adopted unchanged from [1].

Spatial depth (configuration choice): The base paper does not state how many message-passing layers its GNN variants use, so the depth had to be selected here. A single hop gave substantially weaker localization, because most fault locations lie more than one hop from any μPMU (only 6 of 25 fault buses are instrumented); we therefore stack *four* message-passing layers in every GNN variant, fixed for all experiments (Section 4). The depth was selected on RGSAGE-Max alone and then applied to every architecture, so it can favor RGSAGE-Max in the clean-accuracy ranking; Section 5 states which conclusions depend on this choice.

Classification and aggregation: A shared head (batch normalization, ReLU, dropout rate 0.35, linear projection to 26 classes) maps the final embedding of every node to logits z_v. At inference the per-node class probabilities are summed over all N nodes and the arg-max is the graph-level prediction (soft voting, Eq. (5)). Training minimizes the unweighted node-level cross-entropy of Eq. (6), in which each window's event label y supervises all nodes; class balance is handled by the balanced corpus (50% no-fault, with the 25 fault locations equally represented by the $25 \times 11 \times 3$ design) rather than by loss weighting.

$$\hat{y} = \arg \max_{c \in \{0,\ldots,25\}} \sum_{v=1}^{N} \mathrm{softmax}(z_v)_c, \tag{5}$$

$$\mathcal{L} = \frac{1}{N} \sum_{v=1}^{N} \mathrm{CE}(z_v,\, y)\,. \tag{6}$$

3.3. Deviations from the Base Setup

Our setup follows [1] (feeder, μPMU placement, fault taxonomy, windowing, soft-voting evaluation, and an unweighted cross-entropy objective) but differs in four documented respects: a spatial depth of four message-passing hops, which the base paper leaves unspecified; a 1.65 M-window corpus against its 2.5 M; a longer schedule of 30 epochs with gradient clipping and learning-rate warmup, against its 15 epochs for the GNN models and 11 for the GRU; and omission of the reconfigured ("green") topology. All comparisons here use our own corpus and splits, so they are internally consistent and should be read against our own 0.85 reference point rather than the base paper's absolute scale.

3.4. Sensor-Dropout Evaluation Protocol

We evaluate robustness to μPMU failures entirely at *inference* time, with all model parameters fixed, which isolates robustness from any training-time mitigation.

Missingness model: To simulate a lost sensor we select k of the 25 μPMUs and, for every test window, zero all six input channels of each selected node, both the RMS-voltage and the availability-mask channels, which is the encoding of a missing measurement defined in Section 3.1. Section 4.3 also uses a *clustered* policy, in which the k dropped μPMUs form a connected block of the measured-only graph, grown from a uniformly drawn root μPMU by repeatedly adding a uniformly drawn neighbor of the block.

Graph handling: A dropped node keeps its edges and still participates in message passing, but its temporal encoder receives an all-zero sequence and emits a zero-valued embedding that propagates to neighbors, exactly the non-informative message the availability-gated model (Section 3.6) is designed to suppress. All models see the same graph, so differences arise solely from the message-passing mechanism. (An edge-removal policy was implemented but not used, to avoid confounding topology with dropout.)

Dropout sweep and metrics: We sweep $k \in \{0, 1, 2, 5, 10\}$ (0/4/8/20/40% loss), with $k = 0$ the full-observability reference. Dropped sensors are sampled uniformly at random per window, driven by fixed seeds reused across all models so that every architecture sees an identical sequence of dropout realizations on the same frozen test split (Table 2). Each (model, k) point is averaged over $n_{\text{seed}} = 3$ mask realizations and, where stated, over training seeds 42/43/44; the three mask realizations are averaged within each training seed first, so every statistical test in Section 4.5 uses the three seed-level values. Performance is the soft-voted macro-F1 over the 26 classes, giving a macro-F1-versus-k *degradation curve*, from which we derive the *retention* $R(k)$ of Eq. (7) and the *normalized area under the retention curve* (AUDC) of Eq. (8):

$$R(k) = \frac{\mathrm{F1}(k)}{\mathrm{F1}(0)}, \tag{7}$$

$$\mathrm{AUDC} = \frac{1}{k_{\max} - k_{\min}} \int_{k_{\min}}^{k_{\max}} R(k)\, dk \in [0, 1]. \tag{8}$$

Here $k_{\min} = 0$ and $k_{\max} = 10$ are the extremes of the sweep, and Eq. (8) is computed by trapezoidal integration over $k \in \{0, 1, 2, 5, 10\}$; a higher AUDC means slower degradation. Both metrics are relative to a model's own clean score, so they compare how gracefully models degrade, not how accurate they are.

Macro-F1 computation: Each test window receives one graph-level prediction by soft voting over its 25 nodes (Eq. (5)). The predictions of all 247,000 test windows are pooled, without aggregation over the 40 windows of an event, and the per-class F1 scores of the 26 classes (25 fault locations and no-fault) are averaged without weighting, $\mathrm{F1}_{\mathrm{macro}} = \frac{1}{26} \sum_c 2P_cR_c/(P_c + R_c)$, with P_c and R_c the precision and recall of class c; under dropout this is computed for each mask realization and then averaged. The no-fault class is included: it holds 50% of the windows but only 1/26 of the weight in macro-F1, and macro rather than micro averaging is used for this reason, so that the 25 fault classes dominate the score. The node-level macro-F1 of Table 4 instead treats the prediction of each node as a separate sample.

3.5. Dropout-Aware Training

Dropout-aware training removes the train-test mismatch exposed in Section 3.4: the baselines train with all 25 μPMUs available but are evaluated under partial observability. We apply the evaluation missingness model as training-time augmentation, drawing a dropout level $k \sim \mathrm{Uniform}\{0, 1, 2, 5\}$ per batch and zeroing k randomly chosen nodes before each forward pass; the masking random generator is reseeded every epoch, and the k nodes are drawn independently for every window of the batch, with the loss, optimizer, gradient clipping, and warmup of standard training (Table 3). This grid, used for every dropout-aware model except the DA-wide ablation of Section 4.3, does not include $k = 10$, the most severe evaluation level. Because k varies batch to batch, every epoch mixes full- and partial-observability cases, so no μPMU is consistently available and the network cannot depend on any individual sensor. The augmentation touches only the input features, not the graph, architecture, or optimizer, so it applies unchanged to all baselines and to SDR-STGNN (Section 3.6). Masking graph elements during training is itself a known regularizer (e.g., DropEdge [13]); our contribution is not the primitive but framing μPMU dropout as a train-test mismatch and studying its mitigation.

3.6. SDR-STGNN: Availability-Gated Graph Aggregation

SDR-STGNN (Sensor Dropout-Robust Spatio-Temporal Graph Neural Network) addresses sensor dropout architecturally and is included as an ablation against dropout-aware training. All five baselines receive the availability mask as input channels $[m_1, m_2, m_3]$, yet none of the operators in Eqs. (2)–(4) uses that information, so a dropped node keeps contributing its zero-valued embedding as a non-informative message. SDR-STGNN makes the aggregation of the strongest baseline, RGSAGE-Max, *availability-gated*, so dropped sensors send no messages at all.

Availability: For each input window we derive a binary availability indicator for every node from its three availability-mask channels, given by Eq. (9):

$$a_i = \mathbf{1}\left\{ \max_{s} \max_{p \in \{1,2,3\}} m_{i,p}(s) > 0 \right\} \in \{0, 1\}, \tag{9}$$

where $m_{i,p}(s)$ is the mask value of phase p at node i and step s; thus $a_i = 1$ iff any mask channel is active at any step in the window. Every node of the measured-only graph carries a μPMU, so a single- or two-phase unit still yields $a_i = 1$; under the protocol of Section 3.4 the nodes with $a_i = 0$ are exactly those whose sensor has been dropped.

Gated aggregation: Eq. (10) replaces the RGSAGE update of Eq. (3) at each of the four message-passing layers, writing h_i for $h_i^{(l)}$ and h'_j for $h_j^{(l+1)}$: the max-aggregation is restricted to available source nodes, and the self-representation is gated by the same availability indicator:

$$h'_j = \mathrm{ReLU}\Big(W\left[a_j h_j \,\Big\|\, \mathrm{agg}_j\right]\Big), \qquad \mathrm{agg}_j = \max_{\substack{i\in\bar{N}(j)\\ a_i=1}} h_i, \tag{10}$$

where W is the layer projection of Eq. (3) and $\bar{N}(j)$, $\|$, and the inter-hop ReLU convention are as defined there. Nodes with $a_i = 0$ send no messages, and $\mathrm{agg}_j = \mathbf{0}$ when the entire closed neighborhood of j is unavailable. The availability indicators are computed once from the input window (Eq. (9)) and reused across all four layers. The gate adds no parameters, and the shared GRU encoder, four-hop depth, classifier head, and soft voting are identical to RGSAGE-Max; when every sensor is available ($a_i = 1$ for all i), Eq. (10) reduces exactly to the standard RGSAGE-Max update, so the two are identical under full observability ($k = 0$). SDR-STGNN is trained with the dropout-aware augmentation of Section 3.5, so the gates are exercised throughout training. Because everything else is held identical, comparing it against the ungated dropout-aware model separates the training-time augmentation from the structural gate (Section 4).

3.7. Experimental Setup

All models share one training and evaluation configuration; Table 3 lists the training and software settings, and Section 3.4 the evaluation settings. Apart from the variable under study (the message-passing architecture or the augmentation of Section 3.5), every part of the pipeline is fixed, so any difference is attributable to that variable alone. The points not covered by Table 3 are:

- *Batching:* an event-chunk sampler keeps all sliding windows of an event together, consistent with the event-level partition of Table 2.
- *Determinism:* parameter initialization, mini-batch ordering, and the dropout-aware masking are all controlled by one integer seed; no early stopping is used beyond best-validation checkpointing.

Table 3. Training and software configuration shared by all models.

Optimization	
Optimizer	AdamW, learning rate 10^{-3}, weight decay 10^{-4}; batch size 32, 30 epochs
Schedule / clipping	Linear LR warmup over first 3 epochs, constant thereafter; global norm 1.0
Loss / model selection	Node-level unweighted cross-entropy; best validation graph macro-F1
Hardware and software	
GPU	1 × NVIDIA T1000 (8 GB)
Software stack	Python 3.12, PyTorch 2.5.1 (CUDA 12.1), PyTorch Geometric 2.8, OpenDSSDirect.py 0.9.4, NumPy 2.5, scikit-learn 1.9

4. Results

We first establish the baselines under full observability, then quantify how those fixed models degrade as μPMUs are removed at inference (the paper's primary evaluation), then compare dropout-aware training against the availability-gated SDR-STGNN, and finally test whether the training-time fix generalizes across architectures. Unless otherwise

Table 4. Baseline macro-F1 under full observability, as mean±std over seeds 42/43/44, with the best value per column in bold.

Model	Graph-level	Node-level	Macro-F1 by fault resistance		
	Macro-F1	Macro-F1	0.1 Ω	1 Ω	10 Ω
RGSAGE-Max	**0.846 ± 0.007**	**0.718 ± 0.008**	**0.949**	**0.893**	**0.676**
RGSAGE-Mean	0.786 ± 0.023	0.660 ± 0.025	0.906	0.843	0.587
RGCN	0.712 ± 0.017	0.601 ± 0.019	0.874	0.792	0.428
RGATv2	0.522 ± 0.087	0.460 ± 0.080	0.630	0.574	0.312
GRU	0.286 ± 0.009	0.150 ± 0.003	0.410	0.292	0.105

stated, results are computed on the frozen test split of Table 2 and reported as mean ± standard deviation over training seeds 42, 43, and 44, consistent with Section 3.7.

4.1. Baseline Accuracy Under Full Observability

Table 4 summarizes the five baselines under full observability ($k = 0$); the two macro-F1 columns carry ± one standard deviation over seeds 42/43/44, while the per-resistance columns are seed means. The four spatio-temporal GNN models consistently outperform the temporal-only GRU, a gap that shows the importance of spatial message passing here: with only 6 of the 25 fault locations instrumented (Section 3.1), localization usually requires information to propagate to unmeasured buses, for which Eq. (1) alone provides no mechanism. For every model the graph-level macro-F1 exceeds the node-level macro-F1, so the soft vote of Eq. (5) consistently improves on individual node predictions.

Accuracy decreases monotonically as fault resistance increases, for every model: RGSAGE-Max, for example, drops from 0.949 at 0.1 Ω to 0.676 at 10 Ω, reflecting the weaker voltage signatures of high-impedance faults (Section 3.1). Fault detection, in contrast, is effectively saturated: all models achieve a no-fault recall of at least 0.9999 and a no-fault precision of at least 0.972, so the primary challenge is localization rather than detection. Unlike [1], RGATv2 is the weakest GNN model here (0.522 ± 0.087) and the highest in variance across seeds; that spread reflects training instability at the adopted four-layer depth. Stabilizing RGATv2 at that depth is left as future work.

These results also justify the spatial depth of four: a single-seed diagnostic depth sweep on RGSAGE-Max rises monotonically from 0.509 (1 hop, full 30-epoch recipe) to 0.806 (4 hops, reduced budget), with the full 30-epoch recipe reaching 0.846 and no sign of over-smoothing, confirming the enlarged receptive field as the primary driver of accuracy.

4.2. Accuracy Degradation Under Sensor Dropout

We now freeze each baseline and evaluate it under the dropout sweep of Section 3.4. Table 5 summarizes the degradation curves through AUDC (Eq. (8)) and retention at the most severe level ($k = 10$, 40% sensor loss); Fig. 2 shows the full curves. Every model declines as sensors are removed, and no baseline retains more than a third of its full-observability macro-F1 at $k = 10$.

The results reveal a clear trade-off between clean accuracy and robustness; the training-seed standard deviation stays below 0.06 at every k for four of the five models and reaches 0.09 for RGATv2. RGSAGE-Max, the strongest model under full observability, is the least robust, and the inversion is not an artifact of normalizing by each model's own clean score: the curves cross in *absolute* macro-F1 as well. RGATv2's nominal top-robustness position is indicative only, resting on the lowest clean accuracy of the GNNs and a retention spread that overlaps RGCN's, but the inversion does not depend on it: excluding RGATv2 entirely, RGSAGE-Max still retains 5.7% against RGCN's 23.2%. Clean accuracy therefore does not predict robustness to sensor loss, since the most accurate model degrades the fastest. This is consistent with how the aggregation operators treat the zeroed embedding a dropped node propagates (Section 5), and it motivates the two mitigations of Sections 3.5 and 3.6.

Table 5. Degradation under sensor dropout for standard (Std) and dropout-aware (DA) training, as mean ± std over seeds 42/43/44 with three mask realizations each. Bold marks the best *absolute* macro-F1 per column; relative columns are unbolded, normalizing by each model's own clean score.

Model	Standard-training macro-F1 at k					DA macro-F1		AUDC		Ret.@10 (%)	
	k=0	k=1	k=2	k=5	k=10	k=0	k=10	Std	DA	Std	DA
RGATv2	0.522	0.490	0.444	0.302	0.157	0.553	0.505	0.626±0.112	0.961±0.014	31.0±16.4	91.4±2.8
RGCN	0.712	0.664	0.610	**0.418**	**0.164**	0.613	0.533	0.608±0.047	0.940±0.011	23.2± 7.0	87.0±1.9
RGSAGE-Mean	0.786	0.720	**0.647**	0.393	0.108	0.673	0.550	0.541±0.041	0.922±0.034	13.8± 6.4	81.9±8.4
GRU	0.286	0.273	0.252	0.111	0.051	0.300	0.078	0.522±0.081	0.781±0.038	17.9± 4.4	25.4±5.8
RGSAGE-Max	**0.846**	**0.736**	0.590	0.236	0.048	**0.839**	**0.674**	0.402±0.020	0.913±0.006	5.7± 1.8	80.3±1.3

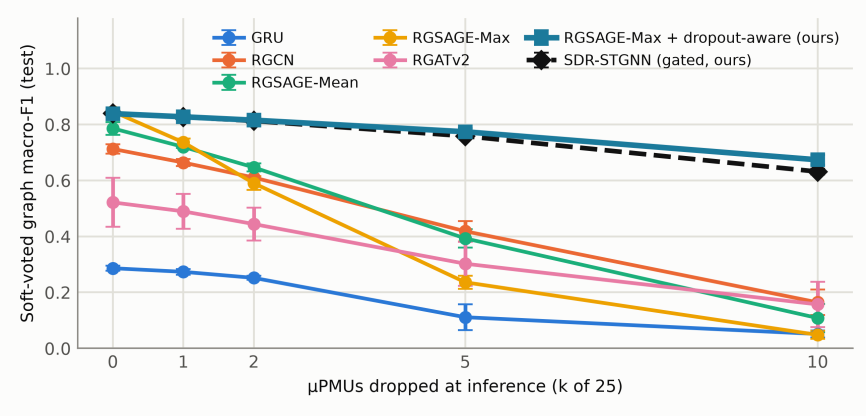


Fig. 2. Soft-voted macro-F1 against the number k of dropped μPMUs, for the five standard-trained baselines and both RGSAGE-Max mitigations: dropout-aware training (blue squares) and the availability-gated SDR-STGNN (black dashed), itself dropout-aware trained. The two nearly coincide — the gate adds no robustness over the augmentation. Error bars are ±1 std over seeds 42/43/44.

4.3. *Effect of Dropout-Aware Training*

Table 6 compares the ways to cope with sensor loss on the identical sweep, all on the RGSAGE-Max backbone: standard training (dropped nodes zeroed), impute-then-infer baselines, dropout-aware training, and the availability-gated SDR-STGNN; the dropout-aware and SDR-STGNN curves are also plotted in Fig. 2.

Imputation baseline: A natural alternative to a training-time fix is to reconstruct dropped sensors before inference. As a lightweight control, the no-training analogue of graph-based imputation [10], we fill each dropped node's three voltage channels with the per-timestep mean of its *available* undirected 1-hop neighbors and present the result to the standard-trained model as observed (mask set to 1, the most favorable setting for imputation); dropped nodes are selected exactly as in the zeroing sweep, so the only change is the fill. This imputer gains little over zeroing (AUDC 0.402 → 0.476; $k = 10$ retention 5.7% → 11.8%, a difference that is not significant at three seeds, Section 4.5) and stays far below dropout-aware training (AUDC 0.913), because the reconstructed values are still fed to a model never trained to distrust them. Unlike the learned pipeline of [10], it needs no advance knowledge of which sensors will fail — exactly what an inference-time failure withholds. To test whether a stronger imputer closes the gap, we also train a learned GNN imputer (Table 6): a masked autoencoder with four residual mean-aggregation (GraphSAGE-style) layers on the measured-only graph, trained on the training split to reconstruct the voltages of $k \sim \text{Uniform}\{1, \ldots, 10\}$ randomly dropped nodes and used exactly like the mean imputer. Its reconstruction error is approximately 18 to 20 times lower (RMSE 0.012–0.017 against 0.24–0.30 p.u.), yet its gains over the mean imputer and over zeroing do not survive Holm correction (minimum $p_{\text{Holm}} = 0.057$ and 0.133), and dropout-aware training exceeds it by 0.486 at $k = 10$ ($p_{\text{Holm}} = 0.037$). One possible explanation is that this error is of the same order as the weakest fault signatures, about 0.01 p.u. at 10 Ω (Section 3.1); the RMSE of fault windows was not measured separately. For 2 of 3 seeds the imputer did not fully converge in 60,000 steps: between steps 55,000 and 60,000 its validation loss changed by −2.3%

for one seed (from 1.02 to 1.00×10^{-3}) and by +4.3% for another (from 0.88 to 0.91×10^{-3}), more than the 2% convergence criterion.

Dropout-aware training raises the AUDC of RGSAGE-Max from 0.402 to 0.913 and its $k = 10$ retention from 5.7% to 80.3%: where the standard model collapses to a macro-F1 of 0.048 at 40% sensor loss, the dropout-aware variant still reaches 0.674 ± 0.017, an absolute gain of 0.626 macro-F1 points. Clean performance is essentially unchanged (0.839 ± 0.011 against 0.846 ± 0.007 at $k = 0$, within one standard deviation and not significant at $p_{\text{Holm}} = 0.196$; Section 4.5), and because the augmentation alters only the training data it adds no parameters and no measurable inference-time cost: measured on one NVIDIA T1000, the standard and dropout-aware checkpoints have identical parameter counts (95,258) and single-event latency (0.639 against 0.640 ms), the augmentation's only overhead being 2.7% on training time per epoch.

Training dropout grid: The grid $\mathcal{K} = \{0, 1, 2, 5\}$ does not include $k = 10$. Adding $k = 10$ to the grid (DA-wide in Table 6; both variants re-evaluated on identical masks[1]) raises macro-F1 at $k = 10$ by 0.030 and lowers clean macro-F1 by 0.006. Both differences have the same sign in all three seeds, but neither survives Holm correction ($p_{\text{Holm}} = 0.057$ each), so we report a consistent trade-off at $n = 3$ and keep $\mathcal{K} = \{0, 1, 2, 5\}$ as the main method.

Targeted (worst-case) dropout: The sweep above removes μPMUs uniformly at random and therefore reports average-case loss. To bound the worst case we repeat it dropping instead the k μPMUs *nearest the fault*, ranked by hop distance over the full 123-bus topology rather than the measured-only graph, on which a fault bus is generally not a node; selection is then a deterministic function of the fault location, so all models see identical masks without a shared seed, and the no-fault half of the corpus, which has no fault bus, retains random selection. Under targeting, standard RGSAGE-Max falls to 0.448 ± 0.039 at $k = 1$ against 0.736 ± 0.013 under random loss, a 3.6× larger drop from its clean score, and to 0.361 against 0.590 at $k = 2$; both differences have the same sign in all three seeds but do not survive Holm correction ($p_{\text{Holm}} = 0.052$ and 0.067), so we report them as consistent observations. From $k \geq 5$ the two policies agree, the standard model having collapsed under either. Dropout-aware training is penalized by targeting at every k (AUDC $0.913 \rightarrow 0.747$, $p_{\text{Holm}} = 0.016$; $p_{\text{Holm}} = 0.021$ at each individual k), so worst-case loss is a distinct stressor rather than one the mitigation absorbs. It nonetheless retains far more than the standard model under the same targeting (+0.307 to +0.473 macro-F1, $p_{\text{Holm}} \leq 0.012$). Its *worst-case* score also exceeds the standard model's *average-case* score at $k = 5$ and $k = 10$ (0.600 against 0.236, $p_{\text{Holm}} = 0.006$; 0.521 against 0.048, $p_{\text{Holm}} = 0.012$); at $k = 2$ the difference (0.695 against 0.590) is consistent but not significant after correction ($p_{\text{Holm}} = 0.085$), and at $k = 1$ the two are indistinguishable (+0.018, $p_{\text{Holm}} = 0.33$). The training-time mitigation therefore transfers to a failure mode it was never trained on.

Persistent (per-event) dropout: A failed μPMU stays dark for the whole event, so we repeat the random sweep drawing the dropped set once per event and holding it across all 40 of its windows. No point on either curve moves by more than 0.003 macro-F1, no difference survives Holm correction across the twelve paired comparisons (min $p_{\text{Holm}} = 0.46$), and the mitigation gain is unchanged (AUDC +0.512, $p_{\text{Holm}} = 0.003$). The reason is that the decision rule is per window: soft voting pools the 25 nodes of one window (Eq. (5)), never the 40 windows of an event, so the prediction for a window depends only on its own mask, whose marginal distribution is the same under both schemes; this equivalence need not hold for an *event-level* rule.

Clustered dropout: Real failures can also be spatially correlated, for example among units that share a communication path. Under the clustered policy of Section 3.4, standard RGSAGE-Max degrades *less* than under random loss (AUDC 0.553 against 0.402; $k = 10$ retention 18.1% against 5.7%), but per k only the difference at $k = 5$ (0.452 against 0.236) survives Holm correction ($p_{\text{Holm}} = 0.023$). For the dropout-aware model the two policies give nearly the same curve (AUDC 0.904 against 0.913; no difference above 0.012 macro-F1, none significant after correction, minimum $p_{\text{Holm}} = 0.065$), and its gain over the standard model under clustered loss is significant at every k (+0.090 to +0.512, $p_{\text{Holm}} \leq 0.022$). We report the milder effect of clustered loss as observed, not established: a connected block does not drop every μPMU equally often, so at $k = 10$ the per-node drop rate ranges from 0.18 to 0.87 against 0.40 under random loss, and the comparison does not separate spatial correlation from which nodes are lost.

The availability-gated SDR-STGNN matches the robustness of dropout-aware training but does not exceed it: its AUDC is 0.892 and its $k = 10$ retention 75.2%, both slightly below the ungated model, and its curve lies below the

[1] Both variants were re-evaluated in one pipeline at a fixed batch size, so that they share mask realizations; the result differs from the published curves by at most 0.0015 macro-F1.

Table 6. Test-time and training-time mitigations on the RGSAGE-Max backbone, as mean ± std over seeds 42/43/44, best per column in bold (the DA-wide ablation row excluded). All variants use the same backbone. Mean and learned imputation have no effect at k=0. DA-wide (training grid $\{0, 1, 2, 5, 10\}$) has a different clean score (0.833), and its relative columns use that score; its values come from a re-evaluation that differs from the published curves by at most 0.0015 macro-F1.

Variant	k=0	k=1	k=2	k=5	k=10	AUDC	Ret.@10 (%)
Standard training (zeroing)	**0.846±0.007**	0.736±0.013	0.590±0.024	0.236±0.023	0.048±0.015	0.402±0.020	5.7±1.8
Mean-imputation (standard)	0.846±0.007	0.739±0.017	0.637±0.028	0.336±0.043	0.099±0.033	0.476±0.042	11.8±4.0
Learned imputation (standard)	0.846±0.007	0.766±0.019	0.686±0.042	0.447±0.087	0.187±0.077	0.570±0.080	22.2±9.3
Dropout-aware training	0.839±0.011	**0.827±0.011**	**0.815±0.011**	**0.774±0.013**	**0.674±0.017**	**0.913±0.006**	**80.3±1.3**
DA-wide (grid with k=10)	0.833±0.013	0.823±0.013	0.812±0.013	0.776±0.013	0.704±0.010	0.928±0.001	84.5±0.1
SDR-STGNN (gated)	0.839±0.002	0.827±0.003	0.813±0.004	0.758±0.006	0.631±0.010	0.892±0.007	75.2±1.3

dropout-aware curve for all $k \geq 1$. At $k = 0$ the gate of Eq. (10) is inactive, so the architecture reduces exactly to its backbone and the two dropout-aware variants score identically (0.839, within seed noise). The gate is also not free: recomputing availability at every hop raises single-event latency by 18% (0.755 against 0.639 ms) for no robustness gain. The structural gate thus adds nothing beyond the training-time augmentation, leaving dropout-aware training the most effective and most practical mitigation evaluated here.

4.4. Dropout-Aware Training Across Architectures

To test whether the gain is specific to the RGSAGE-Max backbone, we retrain every baseline with the identical augmentation and re-evaluate under the same sweep; the resulting values appear in the dropout-aware columns of Table 5. Under random loss, robustness improves for all five architectures, and the AUDC gain is significant for each of them ($p_{\text{Holm}} \leq 0.027$, Section 4.5). The four spatial GNNs rise from AUDC 0.40–0.63 to 0.91–0.96 and from at most 31% to 80–91% retention at $k = 10$. The cost in clean accuracy, however, is not uniform: RGSAGE-Max loses under 0.01 macro-F1 at $k = 0$, but RGCN drops 0.099 (0.712 → 0.613) and RGSAGE-Mean 0.113 (0.786 → 0.673), so the result that robustness comes at no cost to clean accuracy holds for the backbone we adopt rather than for the family as a whole; we did not isolate the cause. Because retention normalizes by each model's own clean score, the relative leaders are again not the strongest in absolute terms: dropout-aware RGSAGE-Max reaches 0.674 macro-F1 at $k = 10$, well above RGATv2's 0.505 despite RGATv2's higher retention. The GNN-free GRU also improves (AUDC 0.52 → 0.78) but its retention stays limited (25%), since without a spatial mechanism it cannot exploit surviving sensors. Robustness recovery is therefore a general property of the training scheme, strongest where the architecture can route information around a failure.

Under clustered loss (Section 4.3), the dropout-aware AUDC is again higher for all five architectures (from 0.517–0.725 to 0.767–0.940; descriptive values). In absolute macro-F1, however, the gain is significant at every k only for RGSAGE-Max ($p_{\text{Holm}} \leq 0.022$); for the other four architectures it is an observation at $n = 3$, the only other significant point being GRU at $k = 5$ ($p_{\text{Holm}} = 0.003$). Under clustered loss, RGSAGE-Mean and RGCN with dropout-aware training also have lower macro-F1 than the standard models at $k = 0$, 1 and 2 (by up to 0.113 and 0.099), as their clean-accuracy cost carries over to light clustered loss.

4.5. Statistical Significance

Every comparison above is a mean over three training seeds, which is a small sample, so we state explicitly which differences survive a test. Table 7 reports paired two-sided t-tests on the seed-wise differences ($n = 3$, df $= 2$) for the seven mitigation claims, Holm–Bonferroni adjusted across the family with the two expected null results included rather than exempted. The seed is the paired unit because every variant is trained from the same three seeds on identical data and splits, so the pairing removes seed-to-seed variation in training difficulty. We do not rely on a Wilcoxon signed-rank test, whose smallest attainable two-sided p-value at $n = 3$ is 0.25. Every other test in this paper follows one rule: each family contains the tests that answer one question (for example, targeted against random loss for one model over the dropout levels), each family is Holm–Bonferroni adjusted on its own, and every reported p_{Holm} is adjusted within its family. For the clustered policy only the per-k comparisons are tested; its AUDC and retention

Table 7. Paired two-sided t-tests over seeds 42/43/44 ($n = 3$, df $= 2$) on the mitigation claims, all on the RGSAGE-Max backbone. Δ is the mean seed-wise difference (F1 points, AUDC, or percentage points for Ret.@10); p_{Holm} is Holm–Bonferroni adjusted over the seven tests. Std = standard, DA = dropout-aware.

Comparison	Δ	95% CI	p	p_{Holm}
DA vs. Std, clean F1 (k=0)	−0.008	[−0.024, +0.009]	0.181	0.196
DA vs. Std, F1 at k=10	+0.626	[+0.557, +0.695]	<0.001	0.004
DA vs. Std, Ret.@10	+74.6	[+67.2, +82.0]	<0.001	0.004
DA vs. Std, AUDC	+0.511	[+0.450, +0.571]	<0.001	0.004
DA vs. imputation, F1 at k=10	+0.574	[+0.449, +0.699]	0.003	0.010
DA vs. learned imputation, F1 at k=10	+0.486	[+0.253, +0.720]	0.012	0.037
DA vs. SDR-STGNN, AUDC	+0.021	[−0.010, +0.052]	0.098	0.196

are reported as descriptive values. The three mask realizations are averaged within each training seed before testing, so the test variance is the training-seed variance and mask realizations are not treated as independent samples; the spread between mask realizations within a seed (0.0002–0.0046 macro-F1) is small against the spread between seeds (0.008–0.061).

The positive claims survive adjustment. Dropout-aware training gains 0.626 macro-F1 at $k = 10$ (95% CI $[0.557, 0.695]$, $p_{\text{Holm}} = 0.004$), 74.6 percentage points of retention and 0.511 AUDC over standard training (both $p_{\text{Holm}} = 0.004$), and exceeds mean-imputation by 0.574 macro-F1 at $k = 10$ ($p_{\text{Holm}} = 0.010$) and the learned imputer by 0.486 ($p_{\text{Holm}} = 0.037$). Extending the same test to every level and every architecture, the gain is significant at each $k \geq 1$ ($p_{\text{Holm}} \leq 0.042$) and the AUDC gain is significant for all five architectures ($p_{\text{Holm}} \leq 0.027$), so the cross-architecture claim of Section 4.4 does not rest on the backbone alone. The two claims we make *as* null results test as null: the clean-accuracy change at $k = 0$ is −0.008 with a CI spanning zero ($p_{\text{Holm}} = 0.196$), and dropout-aware training's 0.021 AUDC edge over SDR-STGNN is not significant ($p_{\text{Holm}} = 0.196$), which is the statistical form of "matches but does not surpass". Mean-imputation's 0.051 gain over zeroing at $k = 10$ is likewise not significant ($p_{\text{Holm}} = 0.307$), sharpening Section 4.3: on this evidence *this* imputer — a training-free neighbor-mean fill — is not distinguishable from leaving the dropped node at zero, a statement about the one control we ran.

Two cross-architecture orderings need an explicit caveat, which reflects the limit of a three-seed design rather than a weakness in the effects themselves. The clean-accuracy ranking of Table 4 reaches only $p_{\text{Holm}} = 0.085$ for its top pair (RGSAGE-Max over RGSAGE-Mean), so no adjacent pair survives Holm adjustment across the four. Neither does the accuracy–robustness inversion: RGCN's absolute macro-F1 at $k = 10$ exceeds RGSAGE-Max's in every one of the three seeds, by a factor of 2.4 to 6.7 ($\Delta = 0.116$), yet $p_{\text{Holm}} = 0.429$, because the *size* of the gap varies with the seed even though its sign does not. A bootstrap over the 6,175 test events gives an interval for the top pair's clean-accuracy gap about five times narrower than the seed-level one, so the number of training seeds, not the size of the test set, bounds these orderings, and we report them as consistent observations at three seeds.

5. Discussion

5.1. Robustness Ranking of the Baseline Architectures

The five baselines expose an inverse relationship between clean accuracy and robustness: the accuracy ranking (RGSAGE-Max > RGSAGE-Mean > RGCN > RGATv2 > GRU) is essentially reversed by the robustness (AUDC) ranking (Tables 4–5). The clearest signal is the inversion at the extremes: the strongest clean model degrades the fastest, retaining only 5.7% at $k = 10$ against RGCN's 23.2%, in the same direction in all three seeds although the three-seed paired test does not reach significance ($p_{\text{Holm}} = 0.429$, Section 4.5). The ordering *among* the mid-tier baselines is indicative rather than established, because the spatial depth was selected on RGSAGE-Max and then applied unchanged to every architecture (Section 4.1); a per-architecture depth could plausibly reorder them, and would in particular raise RGATv2, whose high relative retention rests on a clean score depressed by instability at that depth. These cross-architecture rankings are therefore conditional on the depth of four. The main claims are not: the dropout-aware gain, its cost in clean accuracy, and the comparison with SDR-STGNN are each made within one architecture, between models of identical depth.

This ordering follows from how each operator treats the zeroed embedding a dropped sensor propagates (Section 3.4). The max pooling of Eq. (3) forwards the single largest activation in a neighborhood, so one corrupted message can dominate a live node's update, making RGSAGE-Max the most fragile despite being the strongest clean model, whereas the normalized sum of Eq. (2) and mean pooling average that message against surviving ones and degrade more gracefully. Attention (Eq. (4)) could in principle down-weight a dropped neighbor, but its coefficients are computed from features and are not availability-aware. GRU performs no spatial aggregation, so it is insensitive to neighbors' dropout but capped by the absence of spatial reasoning.

The aggregation operator is therefore a weak lever for robustness, and deployment-level robustness comes from training instead: dropout-aware training, which alters only the input masking, improves robustness on every architecture and recovers most of the loss on the four spatial GNNs (Tables 5 and 6). SDR-STGNN confirms this from the opposite side: silencing every dropped node at every hop also removes the multi-hop paths that would otherwise route information around a failure, so the gate adds nothing over the augmentation. Practically, on this simulated feeder and in RMS phasor simulation, an operator could lose roughly 10 of 25 μPMUs under random, independent loss while macro-F1 stays above 0.67, provided the model is trained with the augmentation; losses concentrated near the fault degrade further (Section 4.3). This suggests that the augmentation can support resilient monitoring of distribution feeders without additional sensing hardware; field validation on real μPMU data is needed to confirm it.

5.2. Why Dropout-Aware Training Works

Separating *that* the augmentation works from *why* matters, because the mechanism determines when it should transfer. The availability-mask channels are static in the clean corpus (Section 3.1), so a network trained on that corpus reads the mask as a fixed per-node fingerprint, never as a live availability signal. Dropping a sensor at inference zeroes all six of its channels, so its mask becomes $(0, 0, 0)$, a pattern that node has never once presented during training: the model has no learned response to it and reads the zeroed voltages as a legitimate near-zero measurement, which the max-aggregation argument above explains will then dominate a live node's update. Dropout-aware training is the first thing that makes the mask vary within a node's own history, so $(0, 0, 0)$ acquires the meaning "this neighbor is dark now" and the network learns the down-weighting for itself rather than being handed it by the architecture.

Three of our results are what that account predicts. First, an explicit architectural gate should then be largely redundant, because the model has already learned the function the gate hard-codes — and SDR-STGNN does not exceed the augmentation (ΔAUDC = 0.021, $p_{\text{Holm}} = 0.196$; Table 7). Second, supplying plausible numbers should not substitute for the training signal, because the deficit is misplaced trust rather than absent values — and mean-imputation, which even sets the mask to 1, gains only 0.051 macro-F1 at $k = 10$ ($p_{\text{Holm}} = 0.307$), while a learned GNN imputer with an approximately 18 to 20 times lower reconstruction error adds only 0.088 more, which is not significant after Holm correction. Third, the augmentation can only exploit redundancy the architecture can reach, so its benefit should scale with spatial connectivity — and the AUDC gain is largest on RGSAGE-Max (+0.511) and smallest on the temporal-only GRU (+0.259), which has no message passing with which to route around a dark node. Dropout-aware training does not reconstruct the missing measurement; it trains the model out of depending on any particular one, and the four-hop receptive field over a 25-node tree is what leaves it somewhere else to look.

5.3. Limitations

Several limitations bound these findings. All experiments use a single feeder, the IEEE 123-bus feeder in its default configuration, so generalization across network sizes and topologies, including the reconfigured ("green") topology of [1] and other standard feeders such as the IEEE 34-bus system, remains open (Section 3.3). The test events are also new load realizations of the same 825 fault scenarios used in training (Section 3.1), so generalization to unseen fault scenarios is not measured. Our main missingness model drops sensors independently and uniformly at random, whereas real μPMU failures are often correlated: co-located units, a shared communication path, or a substation outage fail together, and such structured losses may degrade, and respond to mitigation, differently. Persistence itself is not the gap — holding the dropped set fixed for a whole event leaves every curve unchanged (Section 4.3) — and the clustered policy suggests that connected losses are, if anything, milder for the standard models. That policy, however, changes how often each node is dropped as well as how the dropped nodes are arranged, so the effect of spatial correlation alone remains unmeasured. The degradation curves are consequently averages over random sensor subsets rather than worst

cases; Section 4.3 bounds the worst case by dropping the μPMUs nearest the fault, but only on the RGSAGE-Max backbone and only for proximity-ordered loss, so the other four architectures and other adversarial orderings remain unmeasured. The data are quasi-static RMS phasors: they contain no electromagnetic transients, traveling waves, DC offset, or sub-cycle content, and real-time localization on field μPMU streams is not demonstrated. Voltages are also taken directly from the simulator, without additive measurement noise, quantization, or communication latency, so the full-observability accuracies are optimistic relative to field μPMU data; conversely, the three stable spatial GNNs were still improving at the end of the epoch budget, so their accuracies are a floor. Our imputation comparison covers graph mean-imputation and one learned GNN imputer (Section 4.3), which did not fully converge and was used only with a standard-trained classifier; joint training of the imputer and the classifier was not examined, nor were other learned pipelines such as the Dirichlet/contrastive pipeline of [10] or the VGAE of [6]. The cross-architecture orderings of Tables 4 and 5 are indicative rather than firm, for the reasons established in Section 4.5 and because the spatial depth was selected on RGSAGE-Max alone. Finally, dropout-aware training samples $k \in \{0, 1, 2, 5\}$, so the $k = 10$ evaluation probes beyond the training distribution; recovery remains strong there, and including $k = 10$ in the training grid raises the $k = 10$ result slightly at a small cost in clean accuracy, a trade-off that is consistent across seeds but not significant at $n = 3$ (Section 4.3).

6. Conclusion and Future Work

Building on the STGNN fault-localization methodology of [1], this work contributes a measurement and a negative result. The measurement is that clean accuracy is an unreliable indicator of robustness to μPMU loss: the highest-performing baseline under full observability degraded the fastest as sensors were removed, in absolute as well as relative terms. The negative result is that an availability-gated architecture, SDR-STGNN, achieved comparable but not superior robustness to a purely training-time fix, so on this task the aggregation operator is a weak lever and robustness is better bought during training. That fix, dropout-aware training, raised the retained macro-F1 at 40% sensor dropout from 5.7% to 80.3% on the RGSAGE-Max backbone on the simulated IEEE 123-bus feeder under random, independent loss ($p_{\text{Holm}} = 0.004$) at measured parity in inference latency and parameter count and with no measurable loss of clean accuracy, and, under random loss, improved the robustness of every architecture tested, although on RGCN and RGSAGE-Mean it cost roughly 0.1 macro-F1 under full observability. Future work will investigate spatially correlated sensor failures with per-node drop rates held fixed, worst-case sensor loss beyond the single backbone measured here, larger and more diverse feeder datasets, cross-topology generalization to reconfigured feeders, validation on field μPMU data beyond RMS phasor simulation, joint training of imputers and classifiers, and adaptive availability-gating mechanisms.

Acknowledgements

The authors thank the Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore, Amrita Vishwa Vidyapeetham, for the institutional support that made this work possible.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) in order to draft and edit manuscript text, improve readability, and resolve LaTeX formatting. All experimental design, simulation, training, and evaluation were carried out by the authors, and every reported number, table, and figure was produced by the authors' own experimental pipeline. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Response to Reviewers — Paper ID 1170

Dropout-Aware Training for Sensor-Robust Fault Localization
with Spatio-Temporal Graph Neural Networks

Dear Editor and Reviewers,

We thank you for the careful reading of our manuscript and for the constructive comments. We have revised the manuscript to address every comment. All changed or added text is marked in blue in the revised manuscript. Text that we removed cannot be marked in blue; it is listed in Part F. Page numbers refer to the revised manuscript.

Part A — Reviewer #1

We thank Reviewer #1 for the positive assessment, in particular that "the comparison across five architectures provides useful evidence about sensor-dropout effects," and for the helpful suggestions.

Comment 1.1

> *"Please clarify how the 25 μPMU locations and their topology-derived edges were selected."*

Response. We agree that the description was not precise enough. The 25 μPMU buses are taken unchanged from [1], and we now list them. Each unit measures the phases present at its bus in the feeder model, and the placement is not optimized in this work. We now state the exact edge rule used in our code: on a breadth-first spanning tree of the feeder rooted at the substation (bus 150), each μPMU is connected to the nearest μPMU upstream of it, except that a single- or two-phase unit whose nearest upstream unit is also single- or two-phase is not connected there. Two units (buses 33 and 113) are in that case; a fallback connects them to the μPMU at bus 1 so that the graph stays connected. We state that bus 1 is chosen by a fixed implementation rule (the first connected μPMU in bus-number order), not by electrical proximity. The graph itself is unchanged.

Change. Section 3.1, paragraph 2 (p. 3); Fig. 1 caption (p. 4).

Comment 1.2

> *"Consider evaluating additional feeders or topologies to test generalizability."*

Response. We agree that a single feeder limits generality, and that an evaluation on additional feeders and topologies is the correct next step. This task needs more time than the revision period allows: each new feeder requires a new simulated corpus and the retraining of all five architectures, each with standard and dropout-aware training, at three seeds. We have therefore started the dataset regeneration and training for additional feeders, and we plan to report these results in an extended journal version that continues the same problem statement. In the present manuscript, the single-feeder limitation is now the first item of the Limitations, with concrete next feeders named, and the Conclusion lists cross-topology generalization as future work.

Change. Section 5.3, first item (p. 14); Section 6, future work (p. 15).

Comment 1.3

> *"The conclusions appropriately acknowledge that the architecture rankings are uncertain with only three training seeds."*

Response. We agree, and we kept this caution. In the revision it is stated more precisely: every p-value is now Holm-adjusted within a stated family (Comment 3.2), the cross-architecture rankings are stated to be conditional on the spatial depth (Comment 2.5), and the spread between mask realizations within a seed is shown to be small compared with the spread between seeds (Comment 2.3).

Change. No separate change; see Comments 2.3, 2.5 and 3.2.

Part B — Reviewer #2

We thank Reviewer #2 for the detailed comments, which led to four new experiments.

Comment 2.1

> *"Please test a wider range of training dropout levels, including the evaluation maximum of 10 sensors."*

Response. We added this experiment. We retrained RGSAGE-Max with the training grid $\{0,1,2,5,10\}$ (DA-wide) at three seeds and re-evaluated it together with the original grid $\{0,1,2,5\}$ on identical masks. Adding $k = 10$ to the grid raises macro-F1 at $k = 10$ from 0.675 to 0.704 ($+0.030$) and lowers the clean macro-F1 from 0.839 to 0.833 (-0.006); AUDC rises from 0.913 to 0.928. Both differences have the same sign in all three seeds, but neither is significant after Holm correction ($p_{\text{Holm}} = 0.057$ for each). We therefore report a consistent trade-off at $n = 3$ and keep the original grid as the main method. DA-wide is now a row of Table 6.

Change. Section 3.5 (p. 7); Section 4.3, new paragraph "Training dropout grid" and footnote (p. 11); Table 6, new row (p. 12); Section 5.3 (p. 15).

Comment 2.2

> *"The random sensor-loss protocol should be supplemented with correlated or geographically clustered failures."*

Response. We added a clustered missingness policy: the dropped μPMUs form a connected block of the measured-only graph, grown from a random root μPMU. We evaluated it on all five architectures, for standard and dropout-aware training, at three seeds. For standard RGSAGE-Max, clustered loss is milder than random loss (AUDC 0.553 against 0.402); per dropout level, only the difference at $k = 5$ is significant after Holm correction. For the dropout-aware model the two policies give nearly the same curve, and its gain over standard training under clustered loss is significant at every k ($p_{\text{Holm}} \leq 0.022$). Across the five architectures, the dropout-aware AUDC is higher under clustered loss for all five (descriptive), but the gain in absolute macro-F1 is significant at every k only for RGSAGE-Max. We report the milder effect of clustered loss as observed, not established: a connected block does not drop every μPMU equally often (per-node drop rates from 0.18 to 0.87 at $k = 10$, against 0.40 under random loss), so this comparison does not isolate spatial correlation. We state this in the Limitations and in future work.

Change. Section 3.4, policy definition (p. 6); Section 4.3, new paragraph "Clustered dropout" (p. 11); Section 4.4, new paragraph (p. 12); Section 5.3 (p. 14); Section 6 (p. 15).

Comment 2.3

> *"Clarify whether statistical tests account for the three mask realizations as well as training-seed variation."*

Response. Yes. For each training seed, the three mask realizations are averaged first; the paired t-test then uses the three seed-level values ($df = 2$). Mask realizations are therefore not treated as independent samples. We now state this procedure explicitly, and we report that the spread between mask realizations within a seed (0.0002–0.0046 macro-F1) is small compared with the spread between seeds (0.008–0.061).

Change. Section 3.4, "Dropout sweep and metrics" (p. 6); Section 4.5, paragraph 1 (p. 12).

Comment 2.4

> *"Compare the proposed augmentation with stronger learned imputation methods, not only neighbor-mean imputation."*

Response. We added a learned GNN imputer: a masked autoencoder with four GraphSAGE-style layers on the measured-only graph, trained on the training split to reconstruct the voltages of randomly dropped nodes, and used with the same dropped nodes as the mean imputer. Its reconstruction error is approximately 18 to 20 times lower than that of the mean imputer (RMSE 0.012–0.017 p.u. against 0.24–0.30 p.u.), but its macro-F1 at $k = 10$ is only 0.187, and its gains over mean imputation and over zeroing are not significant after Holm correction. Dropout-aware training exceeds it by 0.486 macro-F1 at $k = 10$ ($p_{\text{Holm}} = 0.037$; new row of Table 7). We report openly that the imputer did not fully converge for two of three seeds, that it was used only with a standard-trained classifier, and that joint training of imputer and classifier was not examined. We did not reimplement the full pipeline of [10], because it assumes that the set of unmeasured buses is fixed and known when the imputer and the classifier are built, which an inference-time failure does not provide.

Change. Section 4.3, imputation paragraph (p. 10); Table 6, new row (p. 12); Table 7, new row (p. 13); Section 5.2 (p. 14); Section 5.3 (p. 15).

Comment 2.5

> *"The four-layer depth was chosen for RGSAGE-Max; discuss its potential effect on comparisons with other architectures."*

Response. We agree. We now state that the depth was selected on RGSAGE-Max alone and then applied to every architecture, so it can favor RGSAGE-Max in the clean-accuracy ranking. We also state that the cross-architecture rankings are conditional on this depth, while the main claims (the dropout-aware gain, its clean-accuracy cost, and the comparison with SDR-STGNN) are each made within one architecture, between models of identical depth. The depth choice is now also listed as a limitation.

Change. Section 3.2, "Spatial depth" (p. 5); Section 5.1 (p. 13); Section 5.3 (p. 15).

Part C — Reviewer #3

We thank Reviewer #3 for the careful comments on the presentation and the statistics.

Comment 3.1

> *"The interpretation of zeroed measurements and availability masks is important and should be emphasized in the methods."*

Response. We added a named paragraph, "Encoding of missing measurements," to Section 3.1. It defines the mask, states that a dropped sensor has all six channels set to zero, explains why a zero voltage alone is ambiguous, and states that the mask is static in the clean corpus (only 14 of the 25 μPMUs are three-phase). Section 3.4 now refers back to this paragraph.

Change. Section 3.1, new paragraph (p. 3); Section 3.4 (p. 6).

Comment 3.2

> *"Please verify the statistical claims and consistently report adjusted p-values where multiple comparisons are made."*

Response. We recomputed every test from the per-seed results. Every p-value in the manuscript is now Holm-adjusted (p_{Holm}) within a stated family, and Section 4.5 states the family rule once: each family contains the tests that answer one question. This changed two conclusions, which we now report faithfully: for standard RGSAGE-Max, the larger damage of targeted (worst-case) loss at $k = 1$ and $k = 2$ is consistent across seeds but not significant after correction ($p_{\text{Holm}} = 0.052$ and 0.067), and the dropout-aware worst-case score exceeds the standard average-case score significantly only at $k = 5$ and $k = 10$. The main claims remain significant (for example, the dropout-aware gain at $k = 10$, $p_{\text{Holm}} = 0.004$). Table 7 now contains seven tests, including the new comparison with the learned imputer, and all its values were recomputed.

Change. Abstract (p. 1); Section 4.3 (p. 11); Section 4.5 (pp. 12–13) and Table 7 (p. 13); Section 5.1 (p. 13); Section 5.2 (p. 14); Section 6 (p. 15).

Comment 3.3

> *"Explain more precisely how the 26-class macro-F1 is calculated from window-level predictions."*

Response. We added a paragraph, "Macro-F1 computation," to Section 3.4. Each test window receives one prediction by soft voting over its 25 nodes; the predictions of all 247,000 test windows are pooled without aggregation over events; and the per-class F1 scores of the 26 classes are averaged without weighting (with the formula). The no-fault class holds 50% of the windows but only $1/26$ of the weight. We also state how the node-level macro-F1 is computed.

Change. Section 3.4, new paragraph (p. 7).

Comment 3.4

> *"The simulation uses RMS phasor data; state this limitation prominently when discussing real-time fault localization."*

Response. We now state the limitation in the Abstract, the Discussion, the Limitations and the Conclusion: the data are quasi-static RMS phasors without electromagnetic transients, traveling waves, DC offset or sub-cycle content, and real-time localization on field μPMU streams is not demonstrated.

Change. Abstract (p. 1); Section 5.1 (p. 14); Section 5.3 (p. 15); Section 6 (p. 15).

Comment 3.5

> *"Some claims about operating with 40% sensor loss should be qualified because results come from one simulated feeder."*

Response. We qualified every such claim: the result holds on the simulated IEEE 123-bus feeder under random, independent loss. We also changed "supports resilient monitoring" to "suggests that the augmentation can support resilient monitoring," and added that field validation is needed.

Change. Abstract (p. 1); Section 5.1 (p. 14); Section 6 (p. 15).

Part D — Editor

> *"Before the submission of the revised manuscript, the English language of your manuscript must be proofread and edited. Strictly follow Procedia Computer Science template."*

Response. English. The authors proofread and edited the English of the full revised manuscript. **Template.** The manuscript uses the Procedia Computer Science LaTeX template (elsarticle with ecrc); the header, footer, margins, captions and equation numbering follow the template. **Similarity.** The similarity index of the revised manuscript is 4% (Turnitin, bibliography excluded); the report is attached as a supplementary file. This letter is included in the same file as the revised manuscript.

Part E — Additional change made by the authors

While revising, we found that the description of the data partition could be read as stronger than it is. The 70/15/15 partition is taken over events, not over fault scenarios: each of the 825 fault scenarios contributes events to the training and test splits, so the test events are new load realizations of the same fault scenarios. We now state this in Section 3.1 and list it as a limitation in Section 5.3 (generalization to unseen fault scenarios is not measured).

Change. Section 3.1 (p. 4); Section 5.3 (p. 14).

Part F — Text removed or shortened (not visible in blue)

To keep the length close to the original despite the added content, we removed or shortened text that repeated other parts of the manuscript. No result and no answer to a reviewer comment was removed.

Location	Change	Where the content remains
Section 3.5, Algorithm 1	Removed	Its specific details (masking random generator reseeded every epoch, nodes drawn per window) are now one sentence in Section 3.5 (p. 7); the per-batch dropout level was already in the text.
Table 3	"Model configuration" and "Evaluation and reporting" rows removed	Sections 3.2 and 3.4; the one value only in Table 3 (RGATv2 attention dropout 0.3) is now in Section 3.2.
Section 3.7	"Objective" bullet removed	Section 3.2 (balanced corpus, no loss weighting).
Section 3.2, paragraph 1	Sentence listing the training method and SDR-STGNN removed	Introduction, paragraph 4.

Location	Change	Where the content remains
Section 3.1, paragraph 4	Opening sentence (“Two encoding choices . . . ”) removed; the paragraph was split	Section 3.1, two paragraphs.
Section 3.4, missingness model	“This reproduces the dataset’s encoding of an absent sensor” replaced by a reference	Section 3.1.
Section 4.3, persistent dropout	Paragraph shortened; individual AUDC and retention values removed	The null result, the test result and the explanation remain.
Section 4.3, imputation and training grid	Numbers moved from the text to Table 6	Table 6.
Section 4.5, bootstrap	Shortened to one sentence	Section 4.5.
Section 5.1, paragraph 3	Repeated numbers removed	Sections 4.2 and 4.3.
Section 5.2, paragraph 1	Description of the mask channels shortened	Section 3.1, “Encoding of missing measurements.”
Section 2	Paragraph 1 shortened; last sentence of the final paragraph removed	Introduction, paragraph 4.
Introduction	Paragraphs 1, 2 and 4 shortened	Sections 2, 3.2 and 3.7, and Table 2.
Fig. 1	Width reduced from 0.70 to 0.60 of the text width	—

We hope that the revised manuscript is now suitable for publication, and we thank the Editor and the Reviewers again for their time.

Sincerely,
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Department of Computer Science and Engineering, Amrita School of Computing, Coimbatore,
Amrita Vishwa Vidyapeetham, India